PAPER 1 / WORKBOOK REVIEW

Expanded explanations and recommended corrections

Questions 1–30 in workbook order. Section and table references follow the November manuscript and its supplements.

Market model and identification

1. State-space realization

Write R = diag(r_i), r_i = 1 - eta_i, and q for trading intensities. The price updates as x_t = a x_(t-1) + kappa q^T y_(t-1) + kappa v_t. Beliefs update as y_t = R y_(t-1) + eta x_t. The state matrix has blocks [a, kappa q^T; a eta, R + kappa eta q^T]. The input vector is kappa(1, eta)^T and the output selects the price. In the pre-update convention the output matrices are CA and CB = kappa. With s = z^(-1), H = kappa/[1 - as - kappa s sum_i q_i eta_i/(1-r_i s)]. The workbook formula follows from the separate negative-feedback equation p = kappa H_plant(u-qp).

Explanation and recommended correction

Substituting the newly updated price into every belief creates the rank-one coupling kappa eta q^T. A change in one cohort can then influence the other cohorts through price. Rewrite Q1 against manuscript equations (1)-(6). If retaining the negative-feedback exercise, specify the price equation, plant numerator, and timing separately.

2. Poles, zeros, and stability

For the workbook plant N/D = sum_i w_i z/(z-eta_i), the closed-loop transfer kappa N/(D+kappa qN) takes value 1/q at each nondegenerate eta_i. Those rates are zeros of sensitivity D/(D+kappa qN). For positive weights the plant is decreasing between its poles, giving one closed-loop pole in (0,eta_1) and one per gap. In the manuscript the numerator is kappa z product_i(z-r_i), giving memory zeros at nondegenerate r_i. Positive-cohort stability is a + kappa sum_i q_i < 1.

Explanation and recommended correction

At z=r_i, clearing fractions leaves a nonzero denominator term from cohort i while the numerator vanishes. At the positive model's stability boundary, a nonzero equilibrium has y_i=x and therefore a+kappa sum_i q_i=1. Keep the workbook interlacing argument separate from the manuscript's stability argument.

3. Pole matching and identification

Normalized weights, rates, and two primitives give 2K+1 free parameters. A rank-K pole map has a local level set of dimension K+1. This establishes pole matching only. Under the manuscript's single-innovation minimum-phase assumptions, the spectrum also identifies the transfer function up to scale, and known primitives identify the merged measure.

Explanation and recommended correction
Two systems can decay at the same rates and have different impact taps or numerators. Rewrite the claim as a limitation of matching poles alone. State the spectral factorization, innovation, and primitive assumptions separately.

4. Merged measures and moments

For common zero initial conditions, equality for all inputs is equivalent to equality of impulse responses. With M(s)=sum_j m_j s^j, invert H to obtain M=(1-as-kappa/H)/(kappa s). Equality of taps h_0 through h_(2K) gives equality of moments m_0 through m_(2K-1). The difference of two K-atomic measures has at most 2K nodes, and the Vandermonde matrix forces every signed weight to be zero.

Explanation and recommended correction
A Lagrange polynomial that is one on a chosen node and zero on the others isolates that node's signed weight. Replace the slow-desk versus fast-HFT example with five desks at two shared rates versus two desks with equal combined weights at those rates. State the finite-atom assumption.

5. Stock and flow roots

If S_t=rho S_(t-1)+u_t and u_t=r u_(t-1)+b R_t, substitution gives S_t-(r+rho)S_(t-1)+r rho S_(t-2)=b R_t. The roots are r and rho, both potentially stable. Changes in holdings are u_t-(1-rho)S_(t-1). The weekly two-root test passed in 2 of 8 markets. See Supplement S18.

Explanation and recommended correction
For r=0.2 and rho=0.95 both roots are stable and swapping their labels leaves the polynomial unchanged. Labeling the learning root needs an ordering assumption or separate flow data. Weekly snapshots aggregate intraperiod flows with retention weights. State the frequency and remove the purported explosive-root argument.

6. Plug-in estimation

The finite tap-to-moment map is smooth when kappa stays away from zero. Baseline demeaning removes constant p*. Root-T errors in taps and primitives preserve the moment error order and the subsequent spacing-dependent rate. From h_0=kappa and h_1=a kappa+kappa^2 m_0 alone, a is confounded with m_0. See Corollary 5.3.

Explanation and recommended correction
Primitive estimation can alter error constants and covariance, while close rates still amplify errors in the second inversion step. Provide a separate identifying source for a or make the corollary conditional on a valid estimator.

Minimax bounds and Prony recovery

7. Moment-matched hard pairs

Take 2K equally spaced nodes with alternating weights proportional to (-1)^j times binomial(2K-1,j). Splitting by sign produces two K-atom measures. Finite differences annihilate degrees 0 through 2K-2, and the first unmatched moment is order Delta^(2K-1). The one-lag feedback makes taps agree through 2K-1.

Explanation and recommended correction
For K=2, use signed weights (1,-3,3,-1). The positive and negative populations each have total mass four before normalization and match the first three moments. State the moment indices and distinguish this construction from the Gauss/Gauss-Radau proof.

8. KL divergence and total variation

For N independent Gaussian resets with covariance sigma^2 I, KL=N times the squared response gap divided by 2 sigma^2. A Taylor expansion on a compact stable rate interval gives a square-summable kernel remainder of order Delta^(2K-1). Uniform stability transfers it to the response. If the squared gap is at most C_K Delta^(4K-2), Pinsker bounds TV by sqrt(C_K N/(4 sigma^2)) Delta^(2K-1).

Explanation and recommended correction
The first terms cancel because of moment matching, and the stable rate ceiling controls late taps. Specify the observed horizon and covariance. With covariance Sigma the per-record squared distance uses the quadratic form Delta h^T Sigma^(-1) Delta h.

9. Le Cam sample requirement

Worst-case error at most alpha requires TV at least 1-2alpha. Therefore N >= 4(1-2alpha)^2 sigma^2 Delta^(-(4K-2))/C_K. Any estimator accurate enough to separate the alternatives yields a test.

Explanation and recommended correction
For the known equal-covariance Gaussian pair with response distance d, projection gives exact minimax error Phi(-sqrt(N)d/(2sigma)). The exact sample threshold is the ceiling of [2sigma Phi^(-1)(1-alpha)/d]^2. Distinguish it from the Le Cam lower-bound constant.

10. Spacing multipliers

Shrinking the gap from 0.10 to 0.01 multiplies the experiment order by 10^6 for two cohorts and 10^10 for three, holding other inputs fixed.

Explanation and recommended correction
A millisecond is a time unit, while the gap measures a difference in fitted memory parameters. A trading-time estimate needs a mapping to memory, a dependence model, and a stationary observation window. Keep the example as a relative experiment multiplier.

11. Prony-Hankel recovery

Deconvolve price taps to obtain 2K moments. Build the K-by-K matrix L_ij=m_(i+j) and solve Lc=-(m_K,...,m_(2K-1))^T. The roots of the monic annihilating polynomial give rates. Solve a Vandermonde system for weights, then use eta_i=1-r_i and q_i=w_i/eta_i.

Explanation and recommended correction
For K=2 the matrix rows are (m_0,m_1) and (m_1,m_2), with right-hand side (-m_2,-m_3). Separate response estimation, feedback inversion, rooting, and weight fitting. Applying Prony directly to price taps estimates closed-loop modes.

12. Hankel conditioning

A 2K-by-2K Hankel matrix from K atoms has rank at most K. For the K-by-K matrix L=V diag(w_i)V^T, positive bounded weights and uniformly separated rescaled nodes give smallest Vandermonde singular value of order Delta^(K-1) and smallest Hankel singular value of order Delta^(2K-2).

Explanation and recommended correction
Factor V into translation, diagonal powers of Delta, and a uniformly invertible rescaled Vandermonde. For K=2 the determinant is w_1 w_2(r_1-r_2)^2. Separate the Hankel exponent 2K-2 from the joint raw-moment inversion exponent 2K-1.

13. Matching upper bound

For tap noise e=sigma/sqrt(N), the largest centered, rescaled moment error is O_p(e Delta^(-(2K-1))). The inverse rescaled moment map is locally uniform. Weight errors have that order and original-rate errors gain one Delta. At weight tolerance epsilon and rate tolerance epsilon Delta, N is of order sigma^2 epsilon^(-2) Delta^(-(4K-2)). See Theorem 5.2.

Explanation and recommended correction
The node conversion is r_i=r_0+Delta s_i. It multiplies the rescaled node error by Delta, while weight errors are unchanged. Replace the workbook's root-perturbation exponent and compare upper and lower bounds using the same accuracy target.

14. ESPRIT and matrix pencil

The hard pair limits every estimator with the same observations. ESPRIT and matrix pencil can differ in finite-sample behavior and numerical conditioning.

Explanation and recommended correction
Separate statistical information from numerical stability. Different methods may improve constants or robustness without changing the minimax exponent. Remove the claim of identical projector conditioning.

Probing, passive data, and disclosure

15. Positive round-trip cost

Positive decreasing taps alone are insufficient. The Toeplitz matrix for (1,0.9,0.1) has determinant -0.468. Positive mixtures of decaying exponentials instead have positive spectral densities, giving a strictly positive quadratic cost for nonzero trades.

Explanation and recommended correction
Buys and sells produce cross terms with opposite signs. Entrywise positivity cannot control the full quadratic form. State the mixture assumption and limit the conclusion to modeled execution costs.

16. Probing energy

A deterministic probe has mean gap (h_1-h_0)*u. If the transfer gap is bounded by L epsilon Delta^(2K-1), KL is bounded by L^2 epsilon^2 Delta^(4K-2) times squared probe energy divided by 2sigma^2. Le Cam yields the reciprocal energy order.

Explanation and recommended correction
Distinguish sum u_t^2 from cumulative absolute quantity sum |u_t|. Derive the impact dependence of L before claiming a particular power of kappa. Converting either norm to dollars requires the execution model.

17. Funded capital barrier

The symmetric known pair has transfer gap O(delta^2). Fixed testing error therefore needs energy of order at least sigma^2 delta^(-4). Coercive quadratic funded execution converts that to a capital lower bound under the stated contract. Halving the separation gives a factor of 16. See Lemma S24.4.

Explanation and recommended correction
The plus and minus perturbations cancel the first-order term, and Gaussian information squares the second-order effect. Remove the claims about evading detection and universal peak inventory scaling. The factor of 16 concerns separation in this benchmark.

18. Annual dollar calibration

Table S10 gives $14.74 million and $356.44 million annually per market at labeled gap Delta eta=0.10, using Roll and one-second noise. Weekly loss figures of about $283,488 and $6,854,612 are multiplied by 52.

Explanation and recommended correction
Define delta=0.05 as the half-gap where used. Label the figures as hypothetical modeled execution losses under weekly renewal. Distinguish the calibration from funded capital and reporting budgets.

19. Passive spectral divergence

The Whittle objective sums log f plus periodogram divided by f. Gaussian KL rate integrates f_0/f_1-1-log(f_0/f_1) with factor 1/(4pi). Uniform spectral ratio perturbation O(Delta^(2K-1)) yields divergence O(Delta^(4K-2)) by a quadratic expansion. A matching order needs a nonzero leading spectral change. See Theorem 5.5.

Explanation and recommended correction

Passive records reveal covariance structure. A transfer change visible only in phase need not change the spectrum. Separate divergence-rate arguments from uniform finite-sample testing control and compare observation counts with reset counts, keeping capital separate.

20. Bitcoin resampling

Multinomial counts with equal probabilities reproduce independent draws with replacement from the empirical shock pool. The reported slopes were -6.01 and -9.81 for synthetic two- and three-cohort signals with real Bitcoin noise. See S16 and Table S23.

Explanation and recommended correction

The experiment retains the empirical marginal tails but loses time ordering. Present fitted slopes with their fitting range or uncertainty if available. A test preserving dependence would address a separate empirical question. Avoid assigning causality to kurtosis.

21. General memory recovery

Moment matching to degree L on [0,rho], rho<1, can keep measures separated by order 1/L while shrinking the response gap geometrically. Choose L proportional to log(N/sigma^2) for the testing bound. Polynomial approximation of Lipschitz functions gives the matching recovery order 1/log(N/sigma^2). See S17.2.

Explanation and recommended correction

The usable degree grows slowly because coefficient growth amplifies noise in higher moments. State the support, observation design, and metric. The manuscript proof uses moment matching and polynomial approximation.

22. Tagged disclosure

Tagged taps f_i,0=w_i h_0 and f_i,1=w_i(h_1+r_i h_0) identify the weight and rate by division. With h_0>0 and weights bounded away from zero, root-N accuracy is independent of spacing. Fixed weight tolerance is constant in spacing and rate tolerance proportional to the gap costs order Delta^(-2).

Explanation and recommended correction

A small weight weakens the rate signal because the formula divides by it. Separate exact cohort-tagged flows, aggregated flows, and holdings. Passive tagged estimation needs its own instrument and noise assumptions.

23. Supervisory choice

With costs C_rep and A Delta^(-p) alone, reporting is cheaper below (A/C_rep)^(1/p). An intermediate band requires an additional option or a gap-dependent value of attribution.

Explanation and recommended correction
Specify the cost of abstention or aggregate intervention and the loss from a wrong targeted decision. A lower boundary exists only if those additional costs or benefits cross reporting cost in the required way.

Empirical findings and policy

24. Eight-market CFTC study

CME Bitcoin, CME Ether, E-mini S&P 500, Treasury notes, euro, yen, WTI and natural gas form the primary sample. The speculative classes are leveraged funds or managed money. Comparators are asset managers or producers/merchants. Tagged intervals were 9.1-71.4 times narrower. H1 passed 8/8, H2 failed at 4/8, and H3 passed 6/8. See Table S25.

Explanation and recommended correction
The extra data directly constrain weekly holdings persistence, which can reflect turnover, reporting aggregation, or inventory retention. Label the fitted parameter accordingly and explain that the fast class was designated by role before estimation.

25. Twelve-market replication

Gold, silver, copper, corn, soybeans, wheat, cattle, coffee, sugar, cotton, Nasdaq-100 and the British pound are the new markets. Tagged widths were 0.014-0.042 versus 0.85-0.96 for prices. H1 passed 12/12, H2 passed 11/12, and H3 failed at 3/12. Table S26 gives coffee about 4,735 times the sample and a 20-market median of 1,353.

Explanation and recommended correction
The equivalent sample multiplier is the squared width ratio under assumed root-T scaling. A width ratio of 23 implies about 529 times the sample. Label this extrapolation and separate it from elapsed time or executed probes. The abbreviated hash 591e71c9 identifies the specification reference. Timing needs separate evidence.

26. Empirical scorecard

Reported passes or holds include CFTC H1 8/8 and H3 6/8, noise K=2 and K=3, two Binance fresh-market checks 30/30, horizon-matched 29/29, tagging advantage 22/29, state tracking, replication H1 12/12 and H2 11/12, and N3 shrinkage 30/30. Failures include CFTC H2 4/8, weekly flow 3/8, two-root 2/8, Binance combined response 5/30, 105-day 10/30, 58-day shrinkage 20/29, return forecast -1.09 points, replication H3 3/12, October N2 1/30, and October N3 5/30. Binance stable learning memory was unsupported. Table S29 also lists N3 sign stability as a failure at 21/30 versus 24 required.

Explanation and recommended correction
The count depends on the treatment of combined and secondary rows. The fresh-market row groups two checks and the N3 sign-stability result adds a failure if counted separately. Give each outcome an ID, primary or secondary status, inclusion flag, threshold, observed result, and disposition. Report failures across both CFTC and Binance.

27. Within-class cancellation

Opposing subgroup signals cancel when their weighted magnitudes nearly match. Distinct exponential rates cannot cancel exactly for every common input with nonzero weights. Failed aggregate tests alone do not identify the mechanism.

Explanation and recommended correction
Two weak responses can look like two strong opposing responses. Distinct rates can also offset early and diverge later. Test the hypothesis with subgroup responses to a common shock, uncertainty intervals, and a comparison to the observed aggregate. A mandatory reporting rule needs its own policy comparison.

28. Spectral blind spot

For triangular A with diagonal eigenvalues lambda_1 and lambda_2 and coupling gamma, the upper off-diagonal of A^t is gamma(lambda_1^t-lambda_2^t)/(lambda_1-lambda_2). At fixed eigenvalues peak gain grows linearly in gamma. Stationary price variance equals s_1/(1-lambda_1^2) plus gamma^2 s_2 times the sum of squared divided differences. Taking s_2 proportional to gamma^(-2) holds stationary variance fixed for positive gamma.

Explanation and recommended correction
s_1 is the fresh price-shock variance conditional on the full prior state. A price-only forecaster can have additional uncertainty about the hidden state. Define the conditioning set and use the positive-coupling family. A diagonal benchmark with the same s_1 has only the first stationary variance term.

29. Pseudospectra and Kreiss bound

The epsilon-pseudospectrum contains eigenvalues and points where the resolvent norm exceeds 1/epsilon. The Kreiss constant is sup over

Explanation and recommended correction
z|>1 of (|z|-1) times the resolvent norm. It lies between peak matrix-power gain divided by en and peak gain. The example bounds what selected summaries can say about transient exposure.|State norms depend on units. A hidden-coordinate rescaling can change numerical state gain while preserving the observable price dynamics when all maps transform consistently. Specify the shock set, units, and loss output before translating gain to market risk. Remove the universal crash multiple.

30. Inventory decisions and targeted halts

For independent tap noise across N resets the finite-horizon sum G has standard error sigma sqrt(H/N). Nonnegative taps and bounded shocks let an upper confidence bound on G size exposure. A targeted halt asks whether removing one cohort crosses a stability threshold, and hard pairs inherit the testing lower bound. See Theorem 0.3 and Proposition 6.4.

Explanation and recommended correction
With signed taps the bounded-input quantity uses the sum of absolute taps. State the stress, horizon, positivity, and observation assumptions. For a halt, define the threshold and decision margin. Two markets on the same side can share an optimal action even when their cohort parameters differ.

Revision priorities

1. Align the workbook with the manuscript equations, rate notation, and theorem numbering.

2. Correct the Hankel dimensions and distinguish numerical conditioning from statistical recovery rates.

3. Separate fitted position persistence from algorithm memory and treat sign cancellation as a hypothesis.

4. State regularity and equilibrium selection in Proposition 3.3. Justify the estimator of a or make Corollary 5.3 explicitly conditional.

5. Reconcile Table S29, gap notation in the dollar calibration, and the definition of one-step variance.

6. Pin the replication code, data instructions, and Lean statements to the manuscript version.

A practical follow-up would compare price-only and class-data rules for a specified intervention on held-out episodes with an independently justified reference outcome.